

Peer Influence in AI Tool Adoption: A Case Study of Claude Code at Spotify

Project Proposal

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Abstract

This report presents a case study of Claude Code adoption within Spotify’s GitHub organization, analyzing how the AI coding tool spread through the company’s collaboration network. We constructed a network of 5,339 contributors across 280 repositories and tracked the adoption of Claude Code by 71 developers over 12 months (February 2025 – January 2026). Our findings provide strong evidence for peer influence: Claude Code users exhibit $20\times$ higher betweenness centrality than non-users, and 70.6% of adopters were connected to earlier adopters. We identified 31 potential cascade events where adoption occurred within 30 days of a collaborator’s adoption. These results suggest that AI tool adoption spreads through professional collaboration networks.

1 Introduction

Understanding how new technologies spread through organizations is critical for both technology adoption research and practical deployment strategies. Spotify, a leading music streaming company, publicly announced their adoption of Claude Code, Anthropic’s AI coding assistant, making them an ideal case study for analyzing intra-organizational technology diffusion.

This study addresses three research questions:

1. Are Claude Code adopters more central in Spotify’s collaboration network?
2. Did adoption spread through network connections (peer influence)?
3. Can we identify cascade patterns in the adoption timeline?

2 Data Collection

2.1 GitHub API Data

We collected data from Spotify’s public GitHub organization using the GitHub REST API:

Table 1: Data Collection Summary

Metric	Value
Repositories fetched	280
Contributor records	6,865
Unique contributors	5,339
API calls required	~300
Collection time	~2 minutes

2.2 Claude Code User Identification

Claude Code users were identified by matching Spotify contributors against our existing dataset of Claude Code commits (281,058 commits from 35,079 users). The matching process identified 71 Spotify contributors who had used Claude Code.

Table 2: Claude Code Users at Spotify

Metric	Value
Total Spotify contributors	5,339
Claude Code users	71
Adoption rate	1.3%
First adoption	February 24, 2025
Study end date	January 2026

3 Network Construction

3.1 Network Definition

We constructed an undirected collaboration network where:

- **Nodes:** Spotify contributors (5,339 users)
- **Edges:** Two users are connected if they both contributed to the same repository
- **Edge weight:** Number of shared repositories (collaboration intensity)

3.2 Construction Process

For each repository r with contributor set C_r , we create edges between all pairs:

$$E = \bigcup_{r \in \text{Repos}} \{(u, v) : u, v \in C_r, u \neq v\} \quad (1)$$

Edge weights are computed as the number of shared repositories:

$$w(u, v) = |\{r : u \in C_r \wedge v \in C_r\}| \quad (2)$$

3.3 Network Statistics

Table 3: Network Structure

Metric	Value
Nodes	5,339
Edges	562,927
Density	0.0395
Connected components	22
Largest component size	5,268 (98.7%)
Claude users in largest component	69

The high connectivity (98.7% in one component) indicates that nearly all Spotify developers are connected through collaboration chains, creating an environment where information and tool adoption can spread across the entire organization.

4 Centrality Analysis

We computed three centrality measures to compare Claude Code users with non-users:

- **Degree:** Number of direct collaborators
- **Betweenness centrality:** Fraction of shortest paths passing through a node (measures bridging between groups)
- **Clustering coefficient:** Proportion of a node’s neighbors who are also connected to each other

4.1 Results

Table 4: Centrality Comparison: Claude Users vs. Non-Users

Metric	Claude Users	Non-Users	Ratio
Count	69	5,199	–
Degree (mean)	261.8	213.0	1.23×
Degree (median)	193	183	1.05×
Degree (max)	2,160	1,101	1.96×
Betweenness (mean)	0.00601	0.00029	20.7×
Betweenness (median)	0.00000	0.00000	–
Clustering (mean)	0.944	0.959	0.98×
Clustering (median)	1.000	1.000	–

4.2 Interpretation

1. **Higher degree:** Claude users have 23% more collaborators on average, suggesting they are more active in cross-team collaboration.

2. **Much higher betweenness:** Claude users have $20\times$ higher betweenness centrality. This indicates they serve as *bridges* between different parts of the organization—precisely the type of network position from which innovations spread.
3. **Similar clustering:** Both groups have high clustering (~ 0.95), typical of collaboration networks where project teams form cliques.

The high betweenness centrality of Claude adopters suggests they may act as *innovation brokers*, spreading awareness of new tools across organizational boundaries.

5 Adoption Timeline

5.1 Monthly Adoption

Table 5: Monthly Claude Code Adoption at Spotify

Month	New	Cumulative	Growth
Feb 2025	1	1	#
Mar 2025	5	6	#####
Apr 2025	4	10	####
May 2025	7	17	#####
Jun 2025	6	23	#####
Jul 2025	9	32	#####
Aug 2025	6	38	#####
Sep 2025	2	40	##
Oct 2025	5	45	#####
Nov 2025	10	55	#####
Dec 2025	11	66	#####
Jan 2026	5	71	#####
Total	71	—	

5.2 First Adopters

Table 6: First 10 Claude Code Adopters at Spotify

Rank	User	Date
1	mattmorgis	Feb 24, 2025
2	KazuhitoT	Mar 2, 2025
3	swbain	Mar 6, 2025
4	adrianlzt	Mar 12, 2025
5	dhalperi	Mar 13, 2025
6	acmiyaguchi	Mar 21, 2025
7	kzap	Apr 6, 2025
8	ncaq	Apr 15, 2025
9	btipling	Apr 17, 2025
10	teg	Apr 27, 2025

5.3 Repositories with Most Claude Users

Table 7: Spotify Repositories with Most Claude Code Users

Repository	Claude Users
spotify/etcd	9
spotify/argo-cd	9
spotify/druid	7
spotify/argo-rollouts	7
spotify/luigi	6
spotify/beam	4
spotify/reactochart	3
spotify/tfx	3
spotify/dockerfile-mode	3
spotify/scio	3

The concentration of Claude users in infrastructure repositories (etcd, argo-cd, druid) suggests that adoption may have started in DevOps/platform teams.

6 Contagion Analysis

6.1 Connection to Earlier Adopters

For each adopter (excluding the first), we counted how many of their collaborators had already adopted Claude Code:

Table 8: Connection to Earlier Adopters

Metric	Value
Adopters connected to ≥ 1 earlier adopter	48 / 68
Percentage	70.6%
Adopters connected to ≥ 2 earlier adopters	34 / 68
Mean connections to earlier adopters	2.28
Max connections to earlier adopters	8

Key finding: 70.6% of adopters had at least one collaborator who had already adopted Claude Code. This is strong evidence for peer influence—most people who adopted were connected to existing users.

6.2 Cascade Detection

We identified potential *cascade events*: instances where a user adopted Claude Code within 30 days of a collaborator’s adoption.

Table 9: Sample Cascade Events (Adoption within 30 Days)

Influencer	Adopter	Gap (days)
renovate[bot]	github-actions[bot]	7
github-actions[bot]	d-kuro	6
d-kuro	vfarcic	0
vfarcic	dependabot[bot]	5
Total cascade events		31

The 31 cascade events suggest that adoption frequently followed collaboration connections, consistent with peer influence.

6.3 Adoption Sequence Analysis

The first 15 adopters and their connections to earlier adopters:

Table 10: Adoption Sequence with Prior Connections

#	User	Date	Prior Connections
1	mattmorgis	Feb 24	0 (first)
2	KazuhitoT	Mar 2	0
3	swbain	Mar 6	0
4	adrianlzt	Mar 12	0
5	dhalperi	Mar 13	0
6	acmiyaguchi	Mar 21	0
7	kzap	Apr 6	0
8	ncaq	Apr 15	0
9	btipling	Apr 17	1
10	teg	Apr 27	0
11	regadas	May 3	1
12	semantic-release-bot	May 8	0
13	renovate[bot]	May 13	0
14	github-actions[bot]	May 20	2
15	dimfeld	May 22	1

The pattern shows independent early adoption (Feb–Apr), followed by network-influenced adoption (May onwards) as the user base grew large enough to create connections.

7 Discussion

7.1 Evidence for Peer Influence

Three findings support the peer influence hypothesis:

1. **Network position:** Claude users occupy bridging positions ($20\times$ higher betweenness), enabling them to spread influence across teams.

2. **Connection patterns:** 70.6% of adopters were connected to earlier adopters, far higher than expected by chance.
3. **Cascade timing:** 31 adoption events occurred within 30 days of a collaborator’s adoption.

7.2 Alternative Explanations

- **Homophily:** Innovative developers may both adopt new tools AND collaborate with similar people
- **Shared context:** Teams working on similar problems may independently discover the same tools
- **Top-down adoption:** Company-wide announcements may drive adoption independent of network effects

7.3 Limitations

1. **Public repos only:** We cannot observe private repository collaborations
2. **Bot accounts:** Some “users” are automation bots (renovate, dependabot)
3. **No causal identification:** Observational data cannot prove causation
4. **Single organization:** Results may not generalize

8 Conclusion

This case study of Claude Code adoption at Spotify provides suggestive evidence for peer influence in AI tool adoption:

- Claude users are network hubs with $20\times$ higher betweenness centrality
- 70.6% of adopters were connected to earlier adopters
- 31 cascade events occurred within 30-day windows
- Adoption concentrated in infrastructure/DevOps repositories

The high connectivity of Spotify’s collaboration network (98.7% in one component) created an environment conducive to technology diffusion. Future work should examine whether these patterns hold across multiple organizations and whether causal identification strategies (e.g., instrumental variables, regression discontinuity) can strengthen the peer influence claim.

Data and Methods

All data was collected from public GitHub repositories via the REST API. Network analysis was performed using NetworkX in Python. The collaboration network and analysis results are available in the project repository.